

Product Review Sentiment Rating Prediction - Executive Summary

Super Team, May 1st, 2026

BOTTOM LINE: We recommend ShopSphere start a 90-day window for pilot deployment of the rating prediction system across different categories. The system would be able to assist the company ahead of the system-wide rollout by identifying negative customer reviews before submission. The model develops the classification performance, and the advantages are review analysis and detecting customer problems. We estimate this technique could reduce 1-to 2-star reviews over time, raising the business performance on seller rankings as well.

BUSINESS IMPACT:

Metric	Current Data	With Model	Change
Review analysis	Manual	Real-time classification	Faster response
Negative review identification	Inconsistent	Early identification	Stable baseline formed
Customer experience	Inconsistent	More structured	Higher visibility

THE APPROACH IN PLAIN ENGLISH: The model analyzes customers' draft reviews before submission. It identifies the patterns in customers' behavior to flag negative experiences potentially early on. By this time, the company would be able to take proactive actions to improve the service before the review is posted.

KEY RISKS:

- **Model Performance Risk:** The model may not detect all reviews all the time, so regular monitoring should be required.
- **Drift Risk:** Customer behavior may change over time, including their vocabulary, which may reduce the model's accuracy. Retraining should be done every 3 months.
- **Adoption Risk:** Training internal teams on how the model works is critical because if they do not use the model's output, the system would be wasted.

RECOMMENDATION Pilot for 90 days: During the pilot, we would track the prediction accuracy in real-time data for a short period, measure the business performance, like response time or detection rate, and review feedback from stakeholders in order to improve the system. After the pilot phase, if performance delivers stable results, then the company should proceed to system-wide deployment and integrate it into the workflow.